TEXORA · LECTURE SHEET

Loom Calculations

A loom is sold by its speed and paid for by its metres. This sheet converts picks per minute into fabric per shift, fabric per shift into looms per order, and stops per hour into the efficiency that connects them — with the weft-consumption calculation that keeps the yarn store honest.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 4 · SPECIALIZATION — FABRIC MANUFACTURING

1. From picks per minute to metres per shift

The loom inserts picks; the customer buys metres. The bridge is the pick density:

$$\text{Picks inserted} = \text{ppm} \times 60 \times \text{hours} \times \text{efficiency}$$

$$\text{Fabric length (m)} = \text{picks inserted} \div \text{picks per metre} = \text{picks} \div (\text{PPI} \times 39.37)$$

WORKED EXAMPLE — PRODUCTION OF ONE LOOM

Loom speed 220 ppm, fabric 60 PPI, efficiency 90%, 8-hour shift.

Picks per hour = 220 × 60 × 0.90 = **11,880**.

Length per hour = 11,880 ÷ 60 = 198 in = 198 ÷ 39.37 = **5.03 m/h**.

Per shift = 5.03 × 8 = **40.2 m**; on three shifts, **120.7 m per loom per day**.

Every element is a lever, and the formula prices each one: +10 ppm at the same efficiency adds 1.8 m/shift; +5 points of efficiency adds 2.2 m/shift — the efficiency lever is usually the cheaper one to pull.

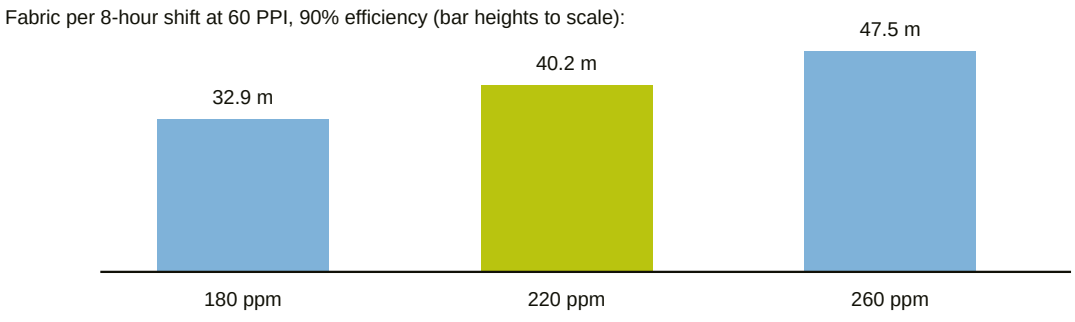


Figure 1 — Output against loom speed for the Section 1 fabric: speed buys metres, but only through the efficiency that survives the shift.

2. Looms per order

WORKED EXAMPLE — ALLOCATING AN ORDER

Order: 50,000 m of the Section 1 fabric, delivery in 12 days, three shifts per day.

Per loom per day = 120.7 m (Section 1).

Looms required = 50,000 ÷ (120.7 × 12) = 34.5 → **35 looms**.

The rounding is not pedantry: 34 looms deliver 49,260 m — 740 m short on the day the truck is booked. Allocation is also made in blocks a weaver can tend and a style can occupy, so the planner's 35 becomes the shed's working number only after the loom-state (which looms are free, on what width, with what shedding motion) has had its say.

3. Efficiency — where it goes

Efficiency is 100% minus the stopped fraction of the shift. If the theoretical running efficiency of the shed is 92% and warp/weft stops average 6 per hour at 1.5 minutes each, the stops alone consume 9 minutes per hour — 15% of the hour — and the realised efficiency falls toward 92 − 15 = **77%** unless stop time is recovered. This is why stop *rate* and stop *duration* are recorded separately: six 1.5-minute stops and one 9-minute stop have the same total but different causes and different cures (yarn quality and settings in the first case; a mechanical fault or a waiting weaver in the second). The production formula of Section 1 is honest only with the realised efficiency inside it.

4. Weft consumption

Weft per metre of fabric = picks per metre × pick length (reed width + waste) × weft tex

WORKED EXAMPLE — WEFT FOR THE SECTION 1 FABRIC

60 PPI → picks per metre = 60 × 39.37 = **2,362 picks**. Reed width 1.70 m (pick length including selvedge waste).

Weft length per metre of fabric = 2,362 × 1.70 = 4,015 m ≈ **4.02 km**.

At Ne 30 (19.68 tex): weight = 4,015 × 19.68 ÷ 1,000 = **79.0 g of weft per metre of fabric**.

Run this per order, not per loom: 50,000 m needs 50,000 × 79.0 g ≈ 3,950 kg of weft yarn — and the winding department (Sheet 1) must have pirns or cones ready at exactly the loom department's rate of eating them.

Glossary

ppm — picks per minute, the loom's speed. **Efficiency** — actual picks inserted as % of the picks the loom would insert running the whole shift. **Pick length** — length of one weft insertion, reed width plus selvedge waste. **Loom allocation** — assigning looms to an order from rate × time. **Stop rate / stop duration** — the two separate halves of lost production.

Key points

• Length = ppm × time × efficiency ÷ picks per metre (worked: 40.2 m/shift; 120.7 m/day on 3 shifts).
• Looms per order = order ÷ (rate × days), always rounded up (worked: 35 looms).
• Stops price themselves: 6 × 1.5 min/h = 15% of the hour.
• Weft per metre = picks/m × pick length × tex (worked: 4.02 km = 79.0 g per metre).

Practice

1. Find the production per shift of a loom at 180 ppm, 52 PPI, 85% efficiency.
2. How many looms, on the Section 1 duty (120.7 m/day), are needed for 80,000 m in 15 days?
3. A fabric at 72 PPI is woven at a reed width of 1.90 m in Ne 24 weft (24.6 tex). Find the weft weight per metre of fabric.
4. A shed's efficiency rises from 85% to 90% at unchanged speed. By what percentage does output rise, and why is this usually cheaper than a speed increase?

Answer key

1. 180 × 60 × 0.85 ÷ 52 ÷ 39.37 × 8 = **35.9 m per shift**.
2. 80,000 ÷ (120.7 × 15) = 44.2 → **45 looms**.
3. Picks/m = 72 × 39.37 = 2,835; length = 2,835 × 1.90 = 5,386 m; weight = 5,386 × 24.6 ÷ 1,000 = **132.5 g/m**.
4. Output scales with efficiency: 90/85 = **+5.9%**. A speed increase costs power, wear, and usually end-breaks (which then eat the gain through Section 3); an efficiency increase is recovered stop time — the picks were already being paid for.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib